

Course syllabus for

Academic year 2025/2026 >

RRY085 - Plasma physics with applications

Plasmafysik med tillämpningar

Course syllabus adopted 2021-02-26 by Head of Programme (or corresponding)

Owner: [MPPHS](#)

7,5 Credits

Grading: TH - Pass with distinction (5), Pass with credit (4), Pass (3), Fail

Education cycle: Second-cycle

Main field of study: Electrical Engineering, Engineering Physics

Department: 70 - SPACE, EARTH AND ENVIRONMENT

Course round 1

Teaching language: English

Application code: 85147

Open for exchange students: Yes

Block schedule: [C](#)

Credit distribution

Module	Sp1	Sp2	Sp3	Sp4	Summer course	No Sp	Examination dates
0108 Examination 7,5 c Grading: TH	7,5 c						Sat 25/10-2025 am J, Thu 08/01-2026 am J, Mon 24/08-2026 pm J

In programs

[MPPHS Physics, Year 1 \(compulsory elective\)](#)

[MPPHS Physics, Year 2 \(elective\)](#)

Examiner:

[Pär Strand](#)

[Go to course homepage](#)

Eligibility

General entry requirements for Master's level (second cycle)

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Specific entry requirements

English 6 (or by other approved means with the equivalent proficiency level)

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Course specific prerequisites

Basic knowledge in electromagnetic field theory, nuclear physics and particle and statistical mechanics.

Aim

The course aims at developing a physical understanding for the characteristic properties of plasmas, including how they can be created and where they appear. An important part of the course is to illustrate plasma physics concepts and phenomena by considering applications ranging from fusion energy generation and microwave techniques to space physics and astrophysics.

Learning outcomes (after completion of the course the student should be able to)

- understand the basic, distinguishing features of plasmas and explain how the plasma state is quantitatively defined in terms of those properties.

- explain the differences between and similarities among the various plasma descriptions presented during the lectures, gain knowledge on what model to apply for a specific problem and roughly estimate their respective validity ranges.

- derive the motion of a single particle in a static and uniform electromagnetic field.
- understand and describe single particle motion in inhomogeneous, temporally varying fields.
- understand the physics underlying Liouville's theorem and explain how it leads to the Vlasov and Boltzmann equations.
- briefly summarize how a fluid model may be constructed from kinetic theory.
- illustrate how to construct a one-fluid model, such as e.g. MHD, from a two-fluid description.
- understand the general mathematical approach used to describe linear plasma waves and apply those methods to new systems (not covered during the lectures).
- derive and interpret plasma wave dispersion relations for the plasma waves covered in the course.
- summarize the physics content of various limits (e.g. high/low frequency limits) of dispersion relations.
- understand the basic mechanism of particle diffusion and what effects it leads to.
- pinpoint the difference between free diffusion, ambipolar diffusion and diffusion in magnetic fields, and explain how to introduce the relevant terms in the one- and two-fluid equations.
- understand how the MHD set of equations are used to determine equilibria and assess stability.
- analyze simple MHD equilibria and stability problems.
- discuss fusion reactions semi-classically in terms of concepts such as cross section, Coulomb and nuclear potentials, energy balance etc.
- understand and recite the reaction chains and the plasma confinement in our Sun.
- discuss and contrast various paths towards controlled fusion energy on Earth.
- describe some typical magnetic confinement devices in terms of geometry, magnetic field structure, stability and technology (e.g. heating and diagnostic systems).
- discuss the various particle orbits in a tokamak and explain the need for a magnetic field twist.
- understand the importance of the fusion triple product and derive the reaction power balance condition.

Content

- Introduction: Definition, occurrence and characteristics of plasmas
- History of plasma physics
- Applications
- Plasma descriptions: Single particle motion, kinetic theory, fluid models
- Waves in plasmas
- Particle diffusion in ionized gases
- Magnetohydrodynamic equilibria and stability
- Fusion energy

Organisation

Lectures

Literature

To be determined

Examination including compulsory elements

Hand-in exercises and oral or written exam

The course examiner may assess individual students in other ways than what is stated above if there are special

reasons for doing so, for example if a student has a decision from Chalmers about disability study support.